



Date : 09/04/2026

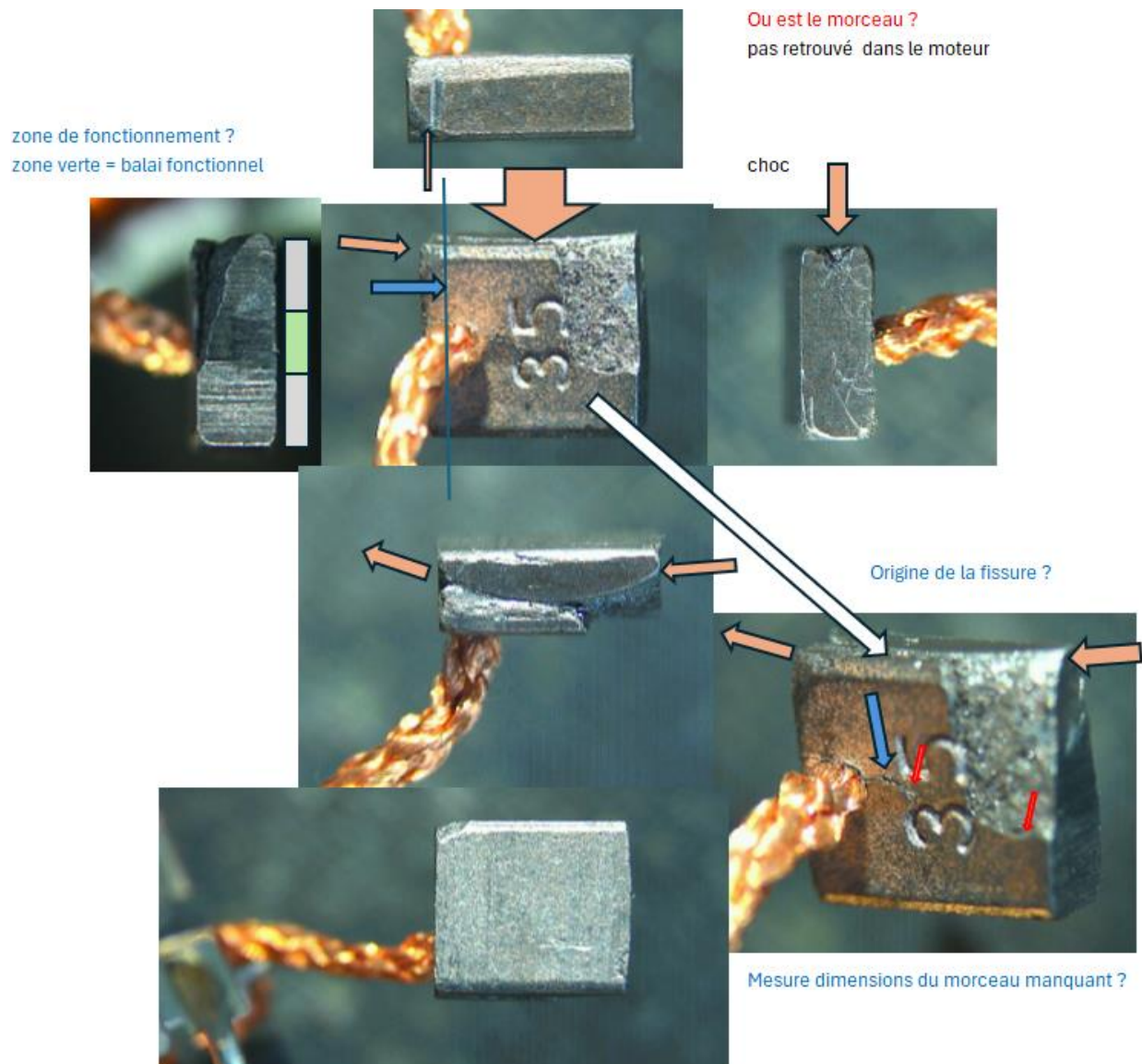
GRADE : 463 00
Ref AVO - 1026649
VALEO – Renault FFK

EXPERT REPORT N° 26-009 :

1. Subject : Return of broken brush from Renault FFK after 1542 km.

2. Pictures with AVOCarbon comment.

➡ Potential constraint orientation



Usinage très plan --> balai probablement endommagé après l'usinage d'épaisseur

From : M.VANHUSE
To : C. BOUVIER
C. LIETARD



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3. Samples :

1 brush card with 3 brushes, one of which is broken.

4. Cracks origin ? :

After polishing at back of brush, there are not crack visible at the cable level.

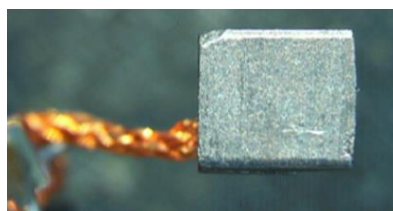


That means that there is no initial cracks in the brush coming from molding.

5. Brush and Damage Dimension:

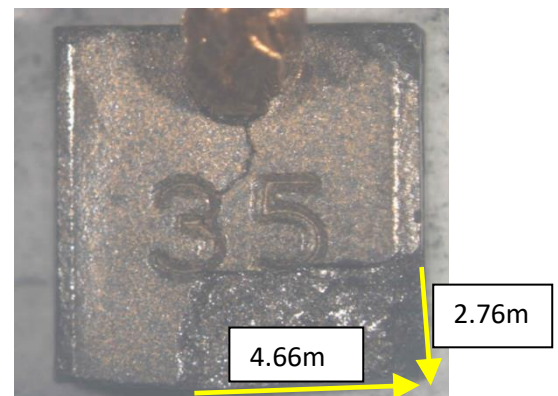
- Total brush thickness: 2.87 mm
The grinded bottom surface is flat and with a good aspect. The dimensions are consistent across the entire piece. If the damage was present at the step of process, it would be not the case.

- Total length: 8.18 mm
- Total width: 7.27 mm



lowest thickness at the level of the chipped: 2.03 mm
Highest thickness at the level of the chipped: 2.51 mm
So missing part has dimensions as follows:

- Average thickness 2.27 x 4.66 x 2.76 mm

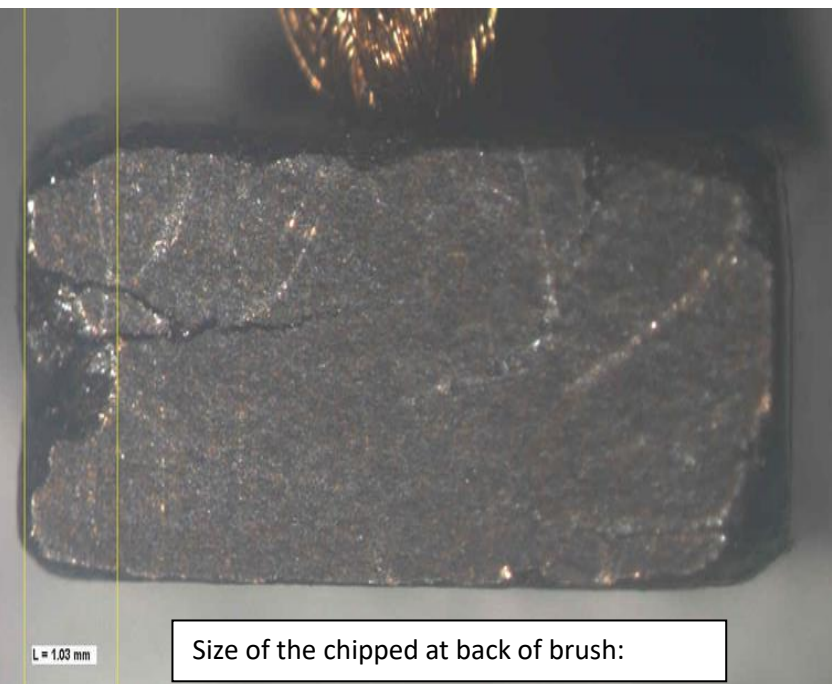
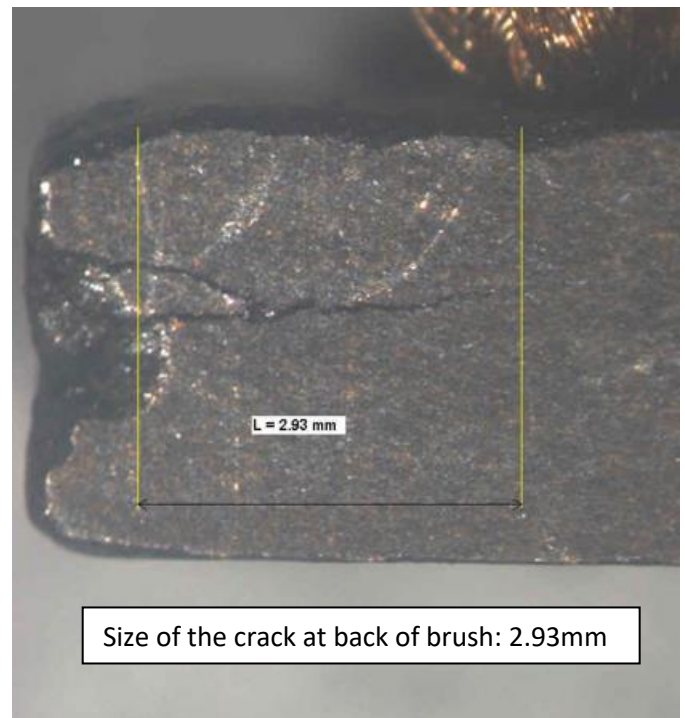
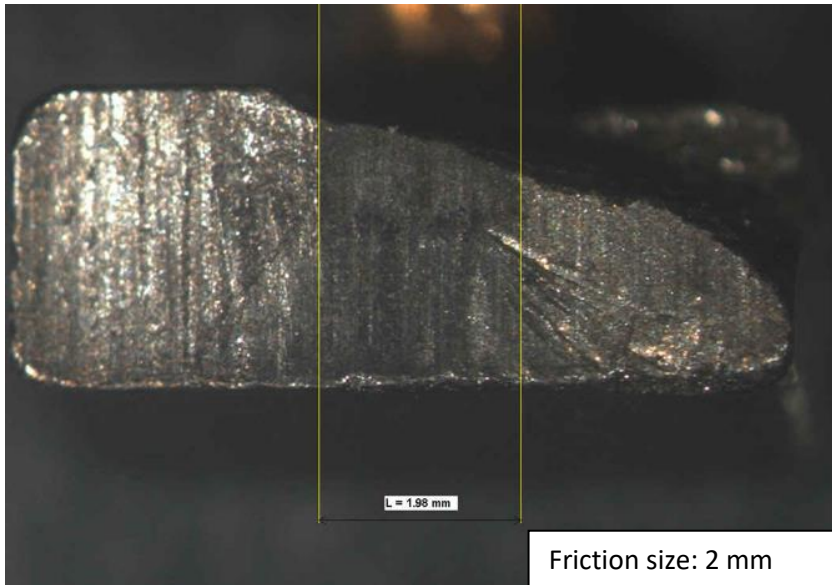




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6. Dimension:





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7. Conclusion:

Analysis of the damaged brush indicates that no damage occurred during the molding process. The damage is characteristic of mechanical stresses, likely occurring after the brush thickness was machined, as this thickness was flat and the correct dimension across the entire part.

Analysis of the round face allows us to conclude that the brush was in contact with the commutator in the correct area (the central third) and was functional despite the damage.

No signs of abnormal operation or excessive temperature were observed.